COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences (NUCES-FAST)

PROGRAM (S) TO BE EVALUATED BS(CS/SE/AI/CY)

A. Course Description

Course Code	CS1005
Course Title	Discrete Structures
Credit Hours	3+0
Prerequisites by Course(s) and Topics	None
Assessment Instruments with Weights (homework, quizzes, midterms, final, assignments etc.)	Midterm examination I: 15% Midterm examination II: 15% End-term examination: 50% Assignments 10% Quizzes: 10% Note: If a student **misses** a **quiz**, it will not be re-conducted. Marks will be deducted for assignments submitted after the due date. Additionally, if the course teacher has published the solution after the due date, then no marks will be awarded for that submission.
Course Coordinator	Mr. Shoaib Raza
URL (if any)	Google Classroom – Link has been provided
Current Catalog Description	Logic, relations, functions, basic set theory, counting, proof techniques, mathematical induction, graph theory, recursion, recurrence relations, number theory, and sequence & series. All the topics will be taught in the perspective of their applications in computing.
Textbook	Kenneth H. Rosen, Discrete Mathematics and Its Applications, McGraw Hill, 8th Edition, 2019.
Reference Material	Sussana S. Epp, Discrete Mathematics with Applications, Brooks Cole, Cengage Learning, 5th Edition, 2020.
Course Goals	A discrete mathematics course has more than one purpose. Students should learn a particular set of mathematical facts and how to apply them; more importantly, such a course should teach students how to think logically and mathematically. To achieve these goals, the focus of this course is on basic mathematical concepts in discrete mathematics and on applications of discrete mathematics in algorithms and data structures. The focus is also on teaching the problem-solving strategies, techniques, and tools and to show students how discrete mathematics can be used in modern computer science. In particular, this course is meant to introduce logic, proofs, sets, relations, functions, counting, and probability, with an emphasis on applications in Computer Science. Further, this course aims to develop understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through study of combinatorial reasoning, abstract algebra, iterative procedures, predicate calculus, tree and graph structures.

A. Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome (CLO)	Domain	Taxonomy Level	PLO	Tools
01	Express statements in terms of predicates, quantifiers and logical connectives.	Cognitive	C2 (Understanding)	4	A1, Q1, M1, F
02	Apply formal logic proofs, logical reasoning to practical problems related to offered program	Cognitive	C3 (Applying)	5	A3, Q3, F
03	Apply mathematical induction to prove properties of sequences, recursive relations.	Cognitive	C3 (Applying)	3	A2, Q2, M2, F
04	Apply graph theory concepts to compute network related metrics and develop solutions for computing applications related to the program.	Cognitive	C3 (Applying)	2	A3, Q3, M2, F

Tool: A = Assignment, Q =Quiz, M = Mid-term, F=Final (End-term)

B. Program Learning Outcomes

For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.

1. Academic Education:	Completion of an accredited program of study designed to prepare graduates as computing professionals.	
2. Knowledge for Solving Computing Problems:	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	✔
3. Problem Analysis:	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	✔
4. Design/ Development of Solutions:	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✔

5. Modern Tool Usage:	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.	✔
6. Individual and Team Work:	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	
7.Communication:	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.	
8. Computing Professionalism and Society:	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.	
9. Ethics:	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.	
10. Life-long Learning:	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.	

C. Relation between CLOs and PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs									
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
CLOs	1				✔						
	2					✔					
	3			✔							
	4		✔								

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	**1. Topics to be covered:**			
	List of Topics	No. of Weeks and Evaluations	Contact Hours	CLO
	The Foundations: Logic and Proofs Introduction Propositional Logic, Applications of Propositional Logic Logical Equivalences Predicates and Quantifiers, Universal and Existential Quantifiers Negating Quantified Statements, Nested Quantifiers, Rules of Inference **Sets, Functions, Sequences and Sums** Sets: Set Operations, Characteristic Vectors: Sets as Bit-Vectors Multisets Functions: Ordered tuples and Cartesian Product, Function and Representations, Types of Functions, Composition and Inverse of Function, Numeric Functions **Relations** Relations and their Properties, Applications of Relations, Representing Relations, Combining Relations, Operations on Relations: Projection and Join Equivalence Relations, and Partial Orderings	**5 (A1 = 3 w, Q1 = 3 w) = 6 w**	**15**	**4,5**
	========== ***MID 1*** ==========			
	Sequences and Sums Sequences and Series **Graphs** Graphs and Graph Models, Terminologies, Types of Graphs, Representing Graphs and Isomorphism, Connectivity, Euler and Hamiltonian Paths, Planar Graphs, and Graph Coloring **Trees** Introduction, Applications, Tree Traversal, Spanning Trees and Minimum Spanning Trees **Introduction to Proofs and Proof Methods**	**5 (A2 = 3 w, Q2 = 3 w) = 6 w**	**15**	**3, 4**

	========== *MID 2* ==========			
	Number Theory and Cryptography Divisibility and Modular Arithmetic, Integer Representation, and Algorithms, Primes and Greatest Common Divisors, Congruences and Applications and Cryptography **Induction and Recursion** Mathematical Induction and Recursive Algorithms **Counting & Counting Techniques** Basics: Introduction and Applications The Sum Rule, The Product Rule The Complement Rule, Inclusion-Exclusion Principle, Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Recurrence Relations.	**5** **(A3 = 4 w, Q3 = 4 w)**	**15**	**2, 3**
	========== ***END-TERM (FINAL) EXAM*** ==========			
	Review	**0.5**	**1**	**2,3,4,5**
	Total	**15**	**45**	

Laboratory Projects/Experients	No Labs and Projects in this course.			
Programming	None			
Class Time Spent on (in contact hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	10	15	20	03
Oral and Written Communications	Students need to participate in class discussion and class assignments.			

Instructor Name _Mr. Shoaib Raza__

Instructor Signature ______________________________

Date: 15-08-2025